

Prediction of Student Exam Performance Using Data Mining Classification Algorithms

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Abstract

- Predict the academic performance of students using machine learning classification algorithms.
- Dataset contains 5,000 student records with academic, behavioural, and demographic attributes.
- Age, Gender, and Department were excluded during feature selection.
- Train-Test Split: 4,000 instances (80%) for training and 1,000 instances (20%) for testing.
- Seven classification algorithms were evaluated: Decision Tree, Random Forest, KNN, Naïve Bayes, SVM, XGBoost, and Logistic Regression.
- Performance was evaluated using Accuracy, Precision, Recall, F-measure, and Confusion Matrix.
- Logistic Regression achieved the best prediction performance.

Dataset Description

- Total records: 5,000 student records with 15 attributes.
- Grade, categorized into five classes A, B, C, D, and F.
- Features: Demographic, academic, and behavioural attributes, including Attendance, Study Hours, Midterm, Assignment, Quiz, Project, Participation, Presentation, Lab, and Final Score.
- Grade breakdown: A = 1,621 (32.42%), B = 958 (19.16%), C = 1,008 (20.16%), D = 755 (15.10%), F = 658 (13.16%).

	Student_ID	Age	Gender	Department	Attendance	Study_Hours_Per_Week	Midterm_Score	Assignment_Score	Quiz_Score	Project_Score	Class_Participation	Presentation_Score	Lab_Score	Final_Score	Grade
0	S1000	28	Male	CSE	87	18	74	71	88	83	9	57	73	88	C
1	S1001	18	Male	EEE	54	33	84	84	98	89	8	94	90	92	A
2	S1002	27	Female	CSE	88	11	81	84	79	81	7	79	83	78	B
3	S1003	19	Female	CSE	82	23	85	83	74	84	7	79	88	81	B
4	S1004	26	Female	English	98	24	95	91	100	89	8	88	93	89	A

Fig. 1 : Sample records from the original student performance dataset.

Methodology

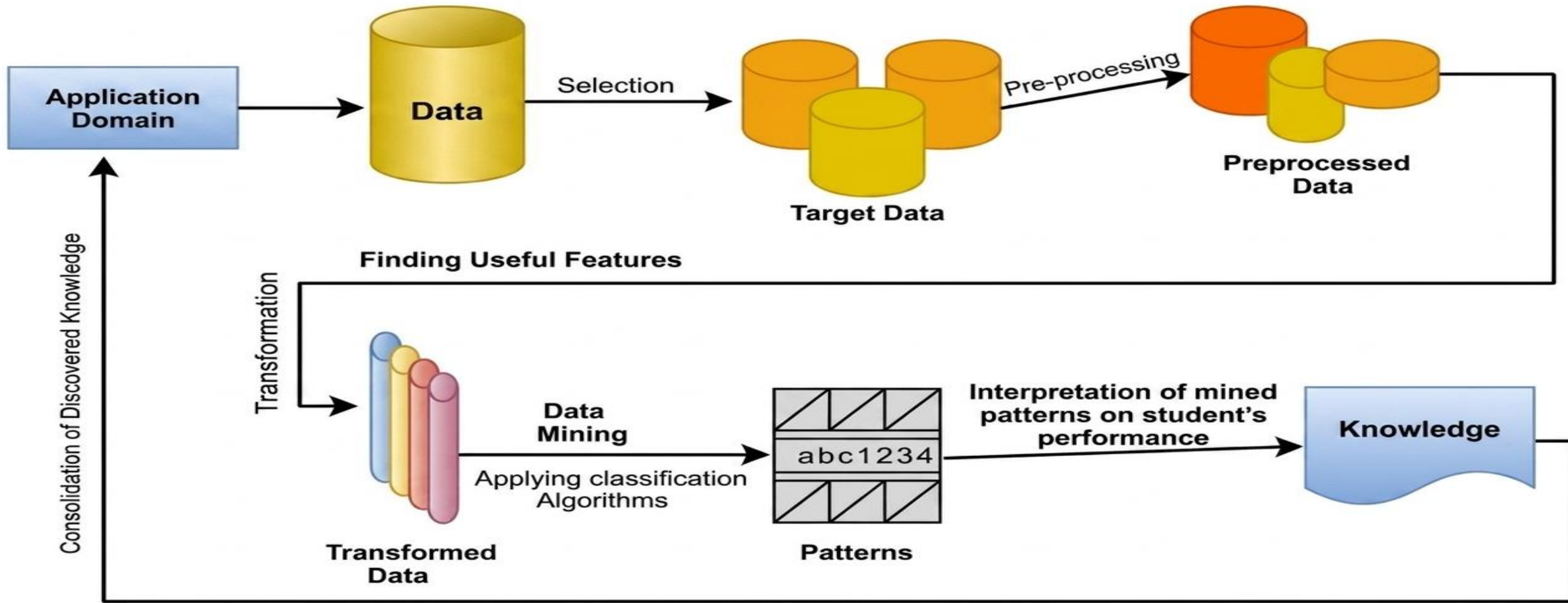
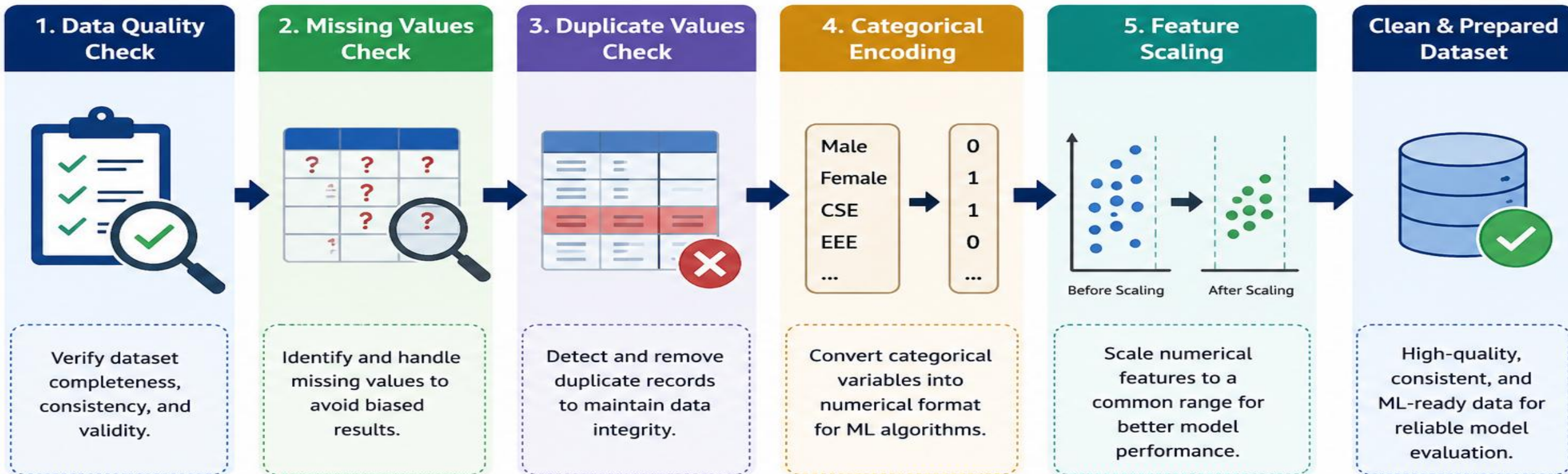


Fig. 2: A diagram for the prediction of student statistics correction exam performance

Pre-processing

Data Preprocessing Pipeline



Purpose: Ensure data quality, maintain consistency, prevent data leakage, and prepare the dataset for reliable machine learning model evaluation.

Feature Selection

- Goal: Identify the most relevant features for predicting student's academic grades and reduce unnecessary feature complexity.
- Target Variable: Student Grade (A, B, C, D, F)
- Low-contribution features: Age, Gender, and Department showed relatively low predictive importance and were excluded from the final model.
- Identifier Removal: Student_ID was excluded because it is only an identifier and has no meaningful predictive value.

Table 1: List of the features of the dataset

Attendance	Study_Hours_Per_Week	Midterm_Score	Assignment_Score	Quiz_Score	Project_Score	Class_Participation	Presentation_Score	Lab_Score	Final_Score
87	18	74	71	66	63	9	57	73	68
54	33	94	84	99	89	6	94	90	92
88	11	81	84	79	81	7	79	83	78
62	23	85	83	74	84	7	79	88	81
96	24	95	91	100	89	8	88	93	89

Feature Selection

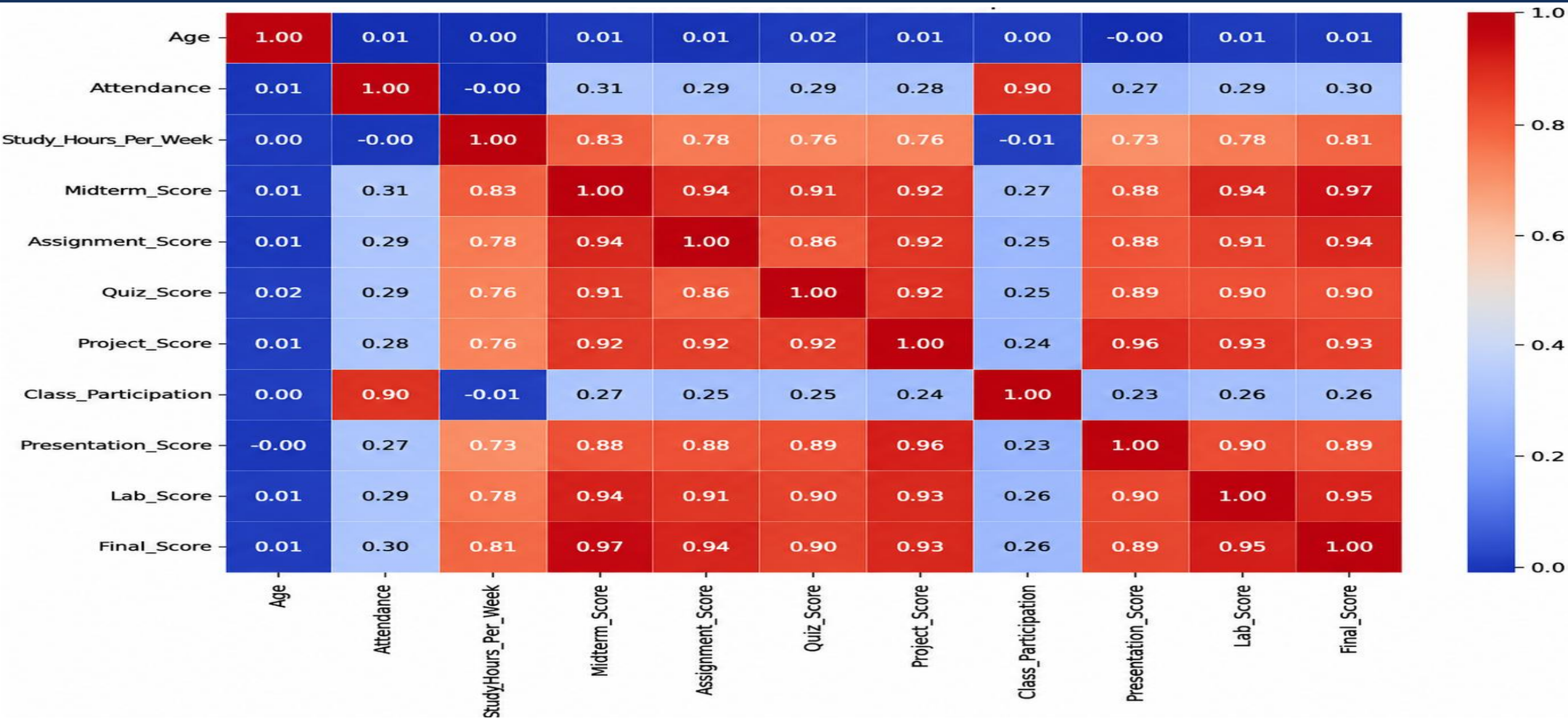
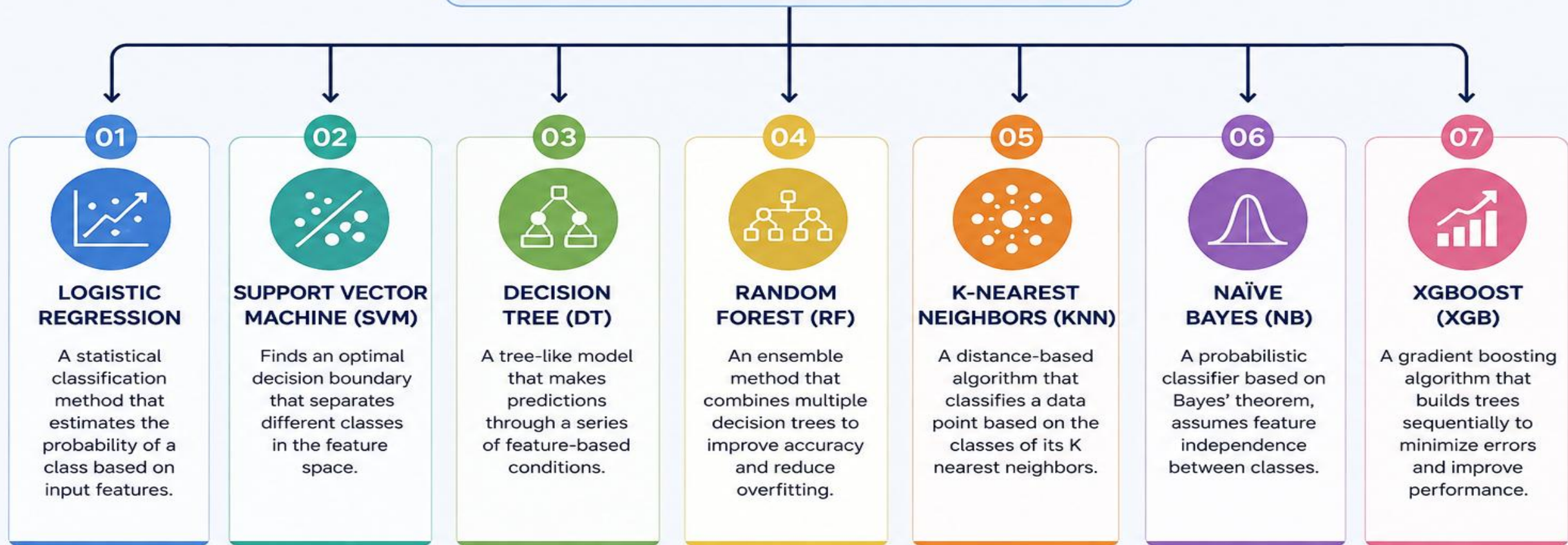


Fig 4: Feature Correlation Heatmap

Machine Learning Classification Algorithms

Seven machine learning classification algorithms were used to predict students' academic grades.



EVALUATION METRICS

Accuracy, Precision, Recall, F1-score, and Confusion Matrix

Experimental Results

Table 2: Overall Performances of all Algorithm

Rank	Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)
1	Logistic Regression	97.9	97.92	97.9	97.90
2	Support Vector Machine	96.9	96.95	96.9	96.90
3	KNN (Optimized)	95.2	95.23	95.2	95.21
4	XGBoost	94.0	94.07	94.0	94.02
5	Random Forest	91.5	91.60	91.5	91.52
6	Decision Tree	87.5	87.56	87.5	87.52
7	Naive Bayes	83.1	83.91	83.1	83.37

Comparison of Experimental Results

Table 3: Performance Comparison Between the Previous Study and the Current Study

Model	Their Table				Our Table			
	Acc.	Prec.	Rec.	F1	Acc.	Prec.	Rec.	F1
Random Forest	98.70	99.00	99.00	99.00	91.50	91.60	91.50	91.52
Decision Tree	98.70	99.00	99.00	99.00	87.50	87.56	87.50	87.52
Naive Bayes	94.00	94.00	94.00	94.00	83.10	83.91	83.10	83.37
Neural Network	96.40	96.00	96.00	96.00	-	-	-	-
KNN	89.60	90.00	90.00	90.00	95.20	95.23	95.20	95.21
Logistic Regression	-	-	-	-	97.90	97.92	97.90	97.90
Support Vector Machine	-	-	-	-	96.90	96.95	96.90	96.90
XGBoost	-	-	-	-	94.00	94.07	94.00	94.02

Logistic Regression- Best Performing Algorithm

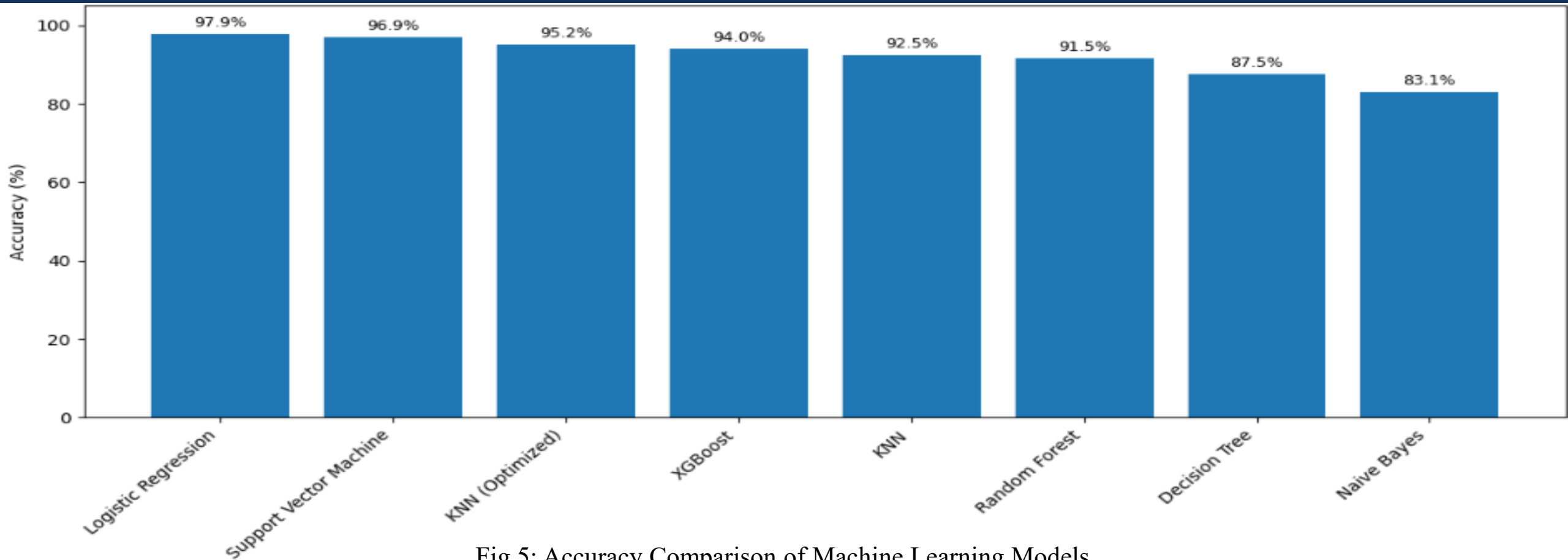


Fig 5: Accuracy Comparison of Machine Learning Models

- The results demonstrate that Logistic Regression provided the most consistent and effective classification performance for predicting students' academic grades in this study.

Logistic Regression- Best Performing Algorithm

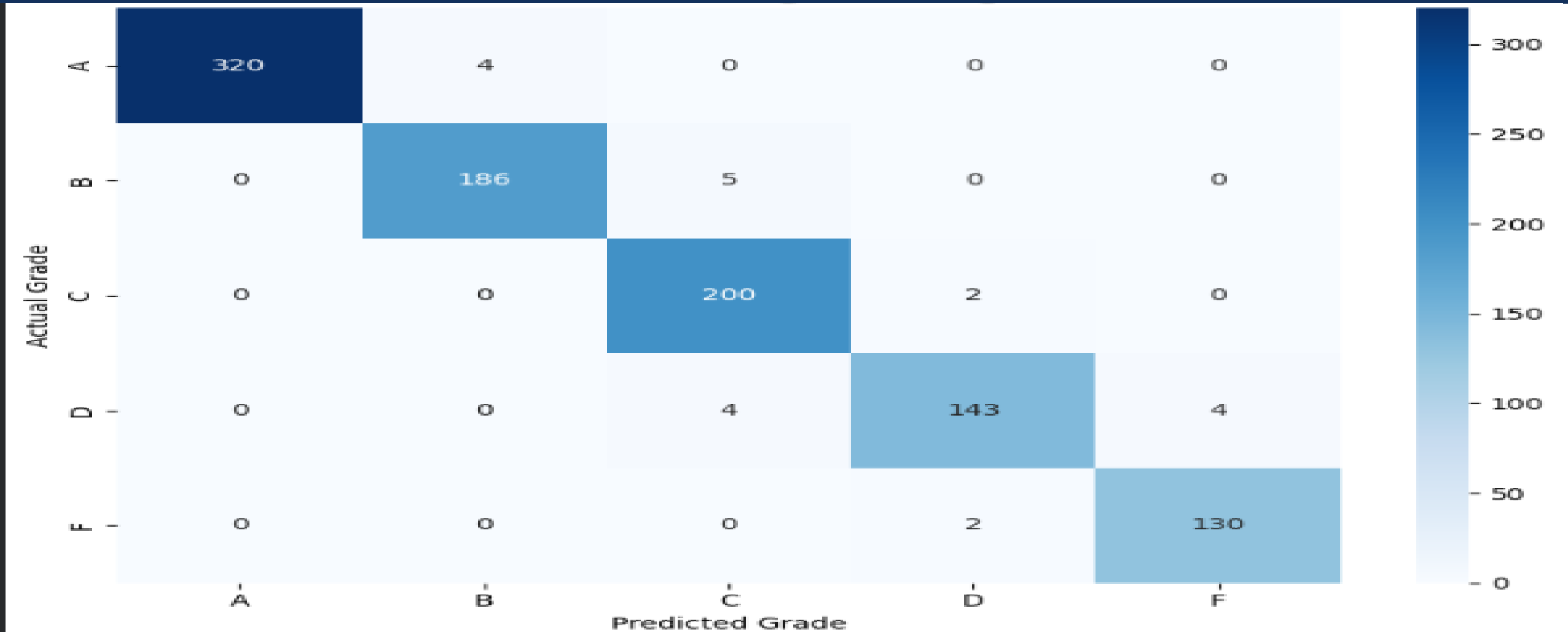


Fig 6: Confusion Matrix - Logistic Regression

Conclusion

- Machine learning can effectively predict student academic performance.
- Logistic Regression achieved the highest prediction accuracy of 97.9%, outperforming the other evaluated models.
- The comparative results demonstrate that model performance varies depending on the dataset and selected features.
- Machine learning can help identify students' academic performance patterns and support timely educational intervention.
- Overall, the study demonstrates the potential of machine learning in educational data mining and student academic grade prediction.
- The findings show that Logistic Regression provided the most effective predictive performance among the seven evaluated algorithms in this study.

Thank You

Questions?

*Prediction of Student Exam Performance Using Data Mining Classification Algorithms
Khairy, Alharbi, Amasha, Areed, Alkhalaf & Abougalala · Education and Information Technologies (2024)*